

SIMATS ENGINEERING

Concept Mapping – CO1-AT2

Course Name:

COMPUTER NETWORKS (CSA-0707)

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Date:

21-07-2026

(Q.1.) Missing Layers in OSI Model

(Concept Map)

Application



Presentation



Session



Transport



Network



Data Link



Physical

Missing Layers

1. Session Layer

2. Data Link Layer

Explanation:

Session Layer

- Establishes communication sessions between devices.
- Maintains synchronization during data exchange.
- Controls dialog between sender and receiver.
- Responsible for session setup, management, and termination.

Data Link Layer

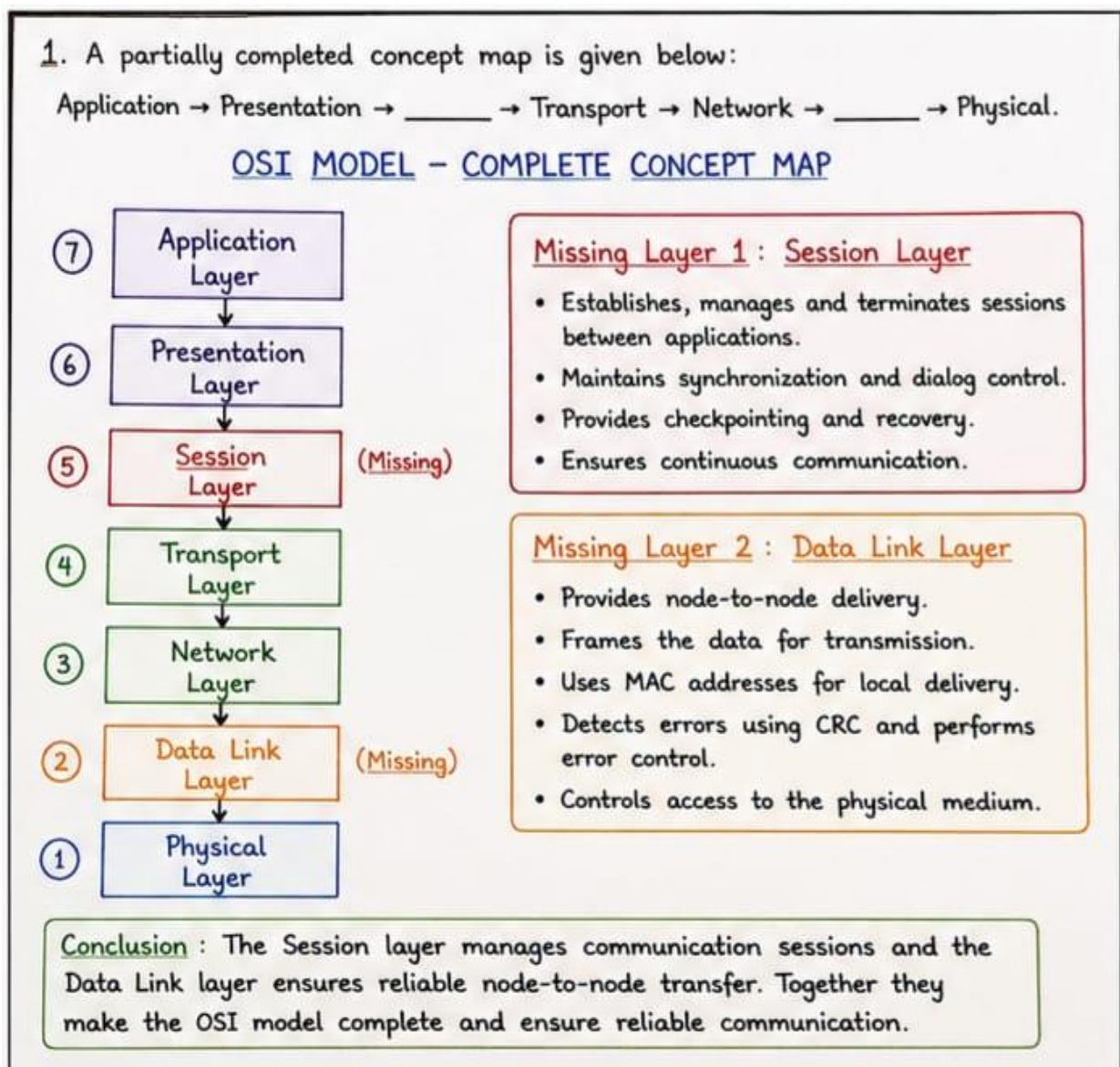
- Provides node-to-node delivery.
- Performs framing of data.
- Detects transmission errors using CRC.
- Uses MAC addresses for local communication.

How They Ensure Reliable Communication

- Session layer maintains continuous communication and recovers interrupted sessions.
- Data Link layer detects transmission errors and ensures correct frame delivery.
- Together with Transport and Network layers, they improve reliability and efficiency.

Conclusion

The complete OSI model consists of seven layers. The Session and Data Link layers play a vital role in maintaining communication and ensuring accurate data transfer, making network communication reliable.



(Q.2.) Concept Map Linking Data Units with OSI Layers

(Concept Map)

Application / Presentation / Session



Data



Transport Layer



Segment



Network Layer



Packet



Data Link Layer



Frame



Physical Layer



Bits

Relationship Between Layers and Data Units

OSI Layer	Data Unit
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Application	Data
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Presentation	Data
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Session	Data
---------	------

Transport	Segment
-----------	---------

Network	Packet
---------	--------

Data Link	Frame
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Physical	Bits
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Transformation

Process:

1. Data → Segment

Transport layer divides data into segments.

Adds port numbers and sequence numbers.

2. Segment → Packet

Network layer adds IP addresses.

Creates packets for routing.

3. Packet → Frame

Data Link layer adds MAC addresses.

Creates frames for local delivery.

4. Frame → Bits

Physical layer converts frames into electrical, optical, or radio signals.

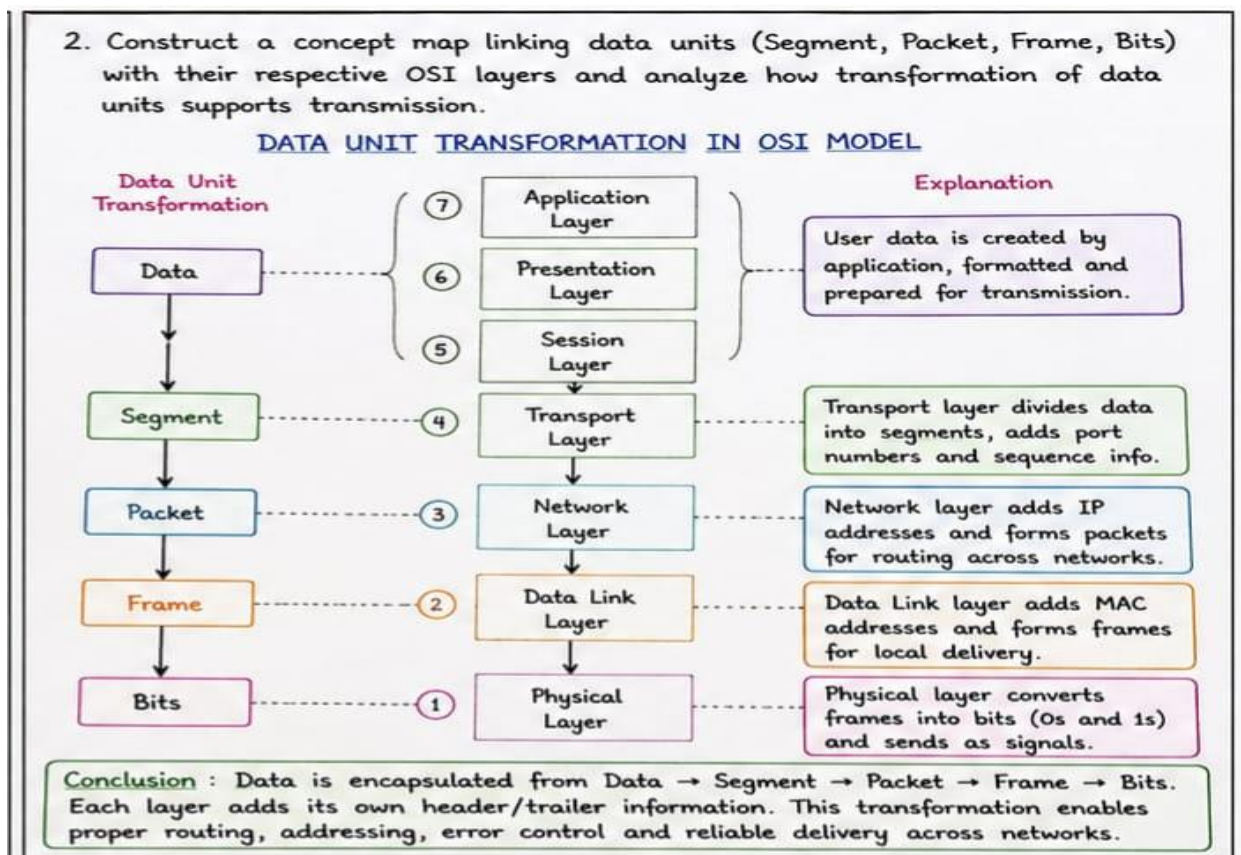
Importance of Transformation

- Enables data to travel through different network layers.
- Supports routing and addressing.

- Provides error detection.
- Ensures efficient and reliable communication.

Conclusion

Data undergoes encapsulation from Data → Segment → Packet → Frame → Bits. Each transformation adds layer-specific information necessary for successful transmission across networks.



(Q.3.)Network Layer and Data Link Layer

(Concept Map)

Network Layer



IP Address



Global Delivery



Destination Network

Data Link Layer



MAC Address



Local Delivery



Destination Device

Explanation

Network Layer → IP Address → Global Delivery

Network layer uses logical addressing (IP Address).

- Determines the destination network.
- Routers use IP addresses to forward packets.
- Enables communication across different networks worldwide.

Example: A packet travels from Chennai to New York using IP addresses.

Data Link Layer → MAC Address → Local Delivery

- Data Link layer uses physical addressing (MAC Address).
- Delivers frames within the local network.

- Switches use MAC addresses for communication.

Example: A frame moves from a laptop to a Wi-Fi router within the same LAN.

How They Work Together

Application Data



Transport



Network (IP Addressing)



Data Link (MAC Addressing)



Physical Transmission

1. Network layer identifies the destination network using IP addresses.

2. Data Link layer identifies the destination device using MAC addresses.

3. Routers handle global delivery.

4. Switches handle local delivery.

Importance for Complete Data Delivery

- IP addresses provide end-to-end communication across networks.
- MAC addresses provide device-to-device communication within a network.
- Together they ensure successful delivery from source device to destination device.

Conclusion

The Network layer ensures global delivery through IP addressing, while the Data Link layer ensures local delivery through MAC addressing. Their combined operation guarantees complete and reliable data transmission across interconnected networks.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

